ASSIGNMENT No. 02

Basic Operations & Expressions: *Write C programs to perform arithmetic, relational, logical, bitwise, and assignment operations. Demonstrate operator precedence and type conversion through suitable examples.*

EXPLORE

Explore some basic header files in C programming like:

<stdio.h> (Standard Input/Output), <stdlib.h> (Standard Library), <string.h> (String Handling), <math.h> (Mathematics), <ctype.h> (Character Handling)

1. WAP to practice math functions such as sin(), cos(), log(), pow(), sqrt(), etc. by including <math.h> header file.
2. WAP to find roots of a quadratic equation (for D>=0 case).
3. WAP to format console output using '\n', '\t', '\b' within the printf statement.
4. WAP to implement assignment operators such as **+=** , **-=** , *=, /=, %=, etc.
5. Assignment operator assigns right hand side value to left hand side variable. Use this idea to interchange (swap) values of two variables. (Hint: You may need a third variable. Think like switching coffee and tea between two cups using a third cup)
6. WAP to shift left and shift right operators (>> and <<). Ask the application of this operator to your lab instructor.
7. WAP to utilize ternary operator (?:).
8. WAP using sizeof() function to find the size of char, integer, long int, float and double.
9. WAP to divide two numbers and use type casting operation (e.g. mean = (double)sum/n;).
10. String is a set of characters (one next to each other stored in the memory and not related to each other), e.g. ="abc123". WAP to practice type casting using the following functions. You can initialize a string using statement char ***s**="3.145", Then **s** is a string.

    /*The **<stdlib.h>** header file in the C programming language provides access to general-purpose utility functions for tasks such as dynamic memory allocation (e.g., malloc, free), process control (e.g., exit, system), numeric conversions (e.g., atoi, atol), and random number generation (e.g., rand, srand). It stands for "standard library" and contains declarations for functions that are not related to input/output operations, which are handled by <stdio.h> */

```
#include <stdlib.h> // Required for atof()
#include <stdio.h>
int main() {
 const char *str1 = "123.45";
 const char *str2 = "-98.76e-2";
 const char *str3 = "invalid_number";
 const char *str4 = " 5.0";
 double num1 = atof(str1);
 double num2 = atof(str2);
 double num3 = atof(str3);
 double num4 = atof(str4);
 printf("Converted value of \"%s\": %f\n", str1, num1);
```

```
printf("Converted value of \"%s\": %f\n", str2, num2);
printf("Converted value of \"%s\": %f\n", str3, num3);
printf("Converted value of \"%s\": %f\n", str4, num4);
return 0;
}
```

11. Write a C program that reads three integers from the user and checks whether the numbers can form the sides of a valid triangle. Use logical operators to apply the triangle condition i.e. “A triangle is valid if sum of any two sides is greater than the third side.” Display a message saying whether the sides form a valid triangle or not.
12. Write a C program using ternary operator to check whether a number is positive, negative, or zero. Take the number from the user at run time.
13. Take an alphabet from the user at run time and check whether it is a vowel or consonant using ternary operator.
14. Using ternary operator, write a program that reads three integers and prints a message if any one of them is equal to the sum of the other two.
15. Write a C program to check whether a given year is a leap year or not using the ternary operator.